



NYU

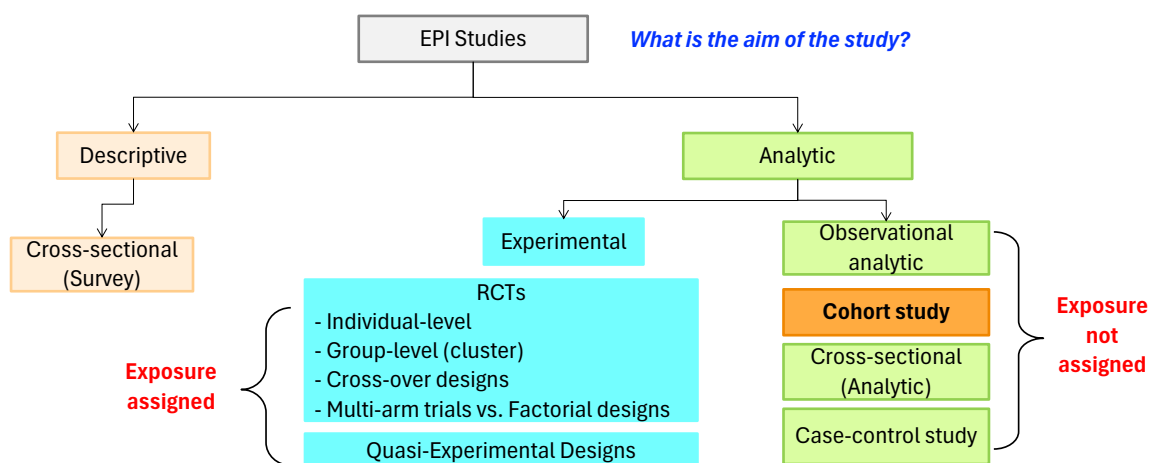
SCHOOL OF GLOBAL
PUBLIC HEALTH

Lecture 7: Cohort Studies

Farzana Kapadia, PhD MPH
Professor of Epidemiology

1

Where are we and why start here?



2

Lecture Objectives

To discuss the following features of cohort studies:

1. Overview of study characteristics
2. Problems and limitations in trials
3. Biases

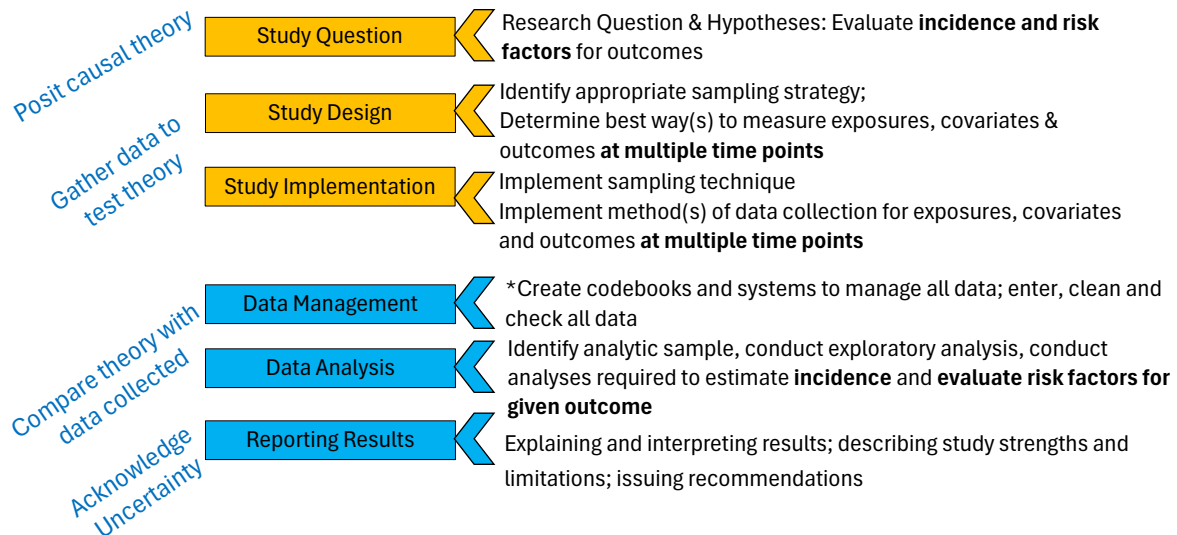
3

Descriptive vs. Analytic Epidemiology

- **Descriptive Epidemiology:**
 - describing differences in disease distribution across populations
 - identifying associations with other variables over which the distribution of disease differs
- **Analytic Epidemiology:**
 - Causal research to understanding the extent to which changes in the exposure cause changes in the outcome on average in a population
 - We often frame studies as “risk factor” studies but do not provide clear definition of what the risk factor is, how it operates (read: need a DAG!)
 - “What is the prevalence of lung cancer among smokers?” could be clearly framed as a descriptive question
 - “Does smoking increase the risk of developing lung cancer?” is a causal question where appropriate confounder control would be required

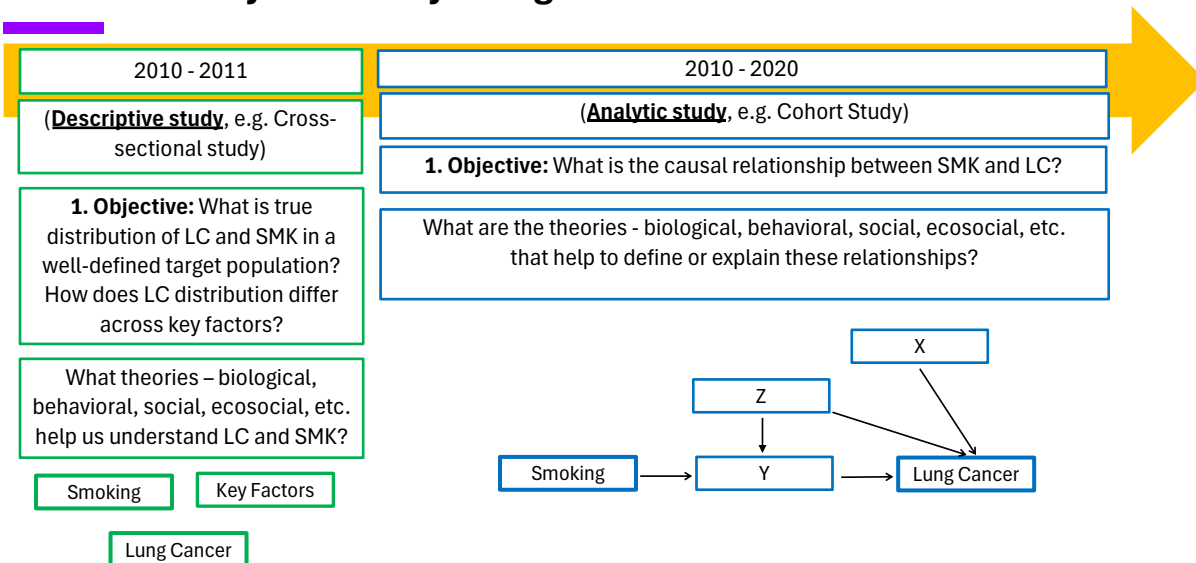
4

Aligning concepts to methods for cohort studies



5

Causal Theory and Study Designs



Lesko CR et al. A Framework for Descriptive Epidemiology. AJE 2022;191(12):2063-2070.

6

Research Questions and Study Designs

2010 - 2011	2010 - 2020
(Descriptive study, e.g. Cross-sectional study) 2. Research Question: <i>“What is prevalence of LC among SMK and ~SMK?”</i> 3. Measures: Lung cancer - self-reported LC (Y/N) - Date of physician diagnosis of LC - EHR record data of LC - Type of LC test (CT) Smoking: - Packs of cig/day in last yr - # of years smoking - Ever/current/etc. - Type of tobacco product	(Analytic study, e.g. Cohort Study) 2. Research Question: <i>“Does smoking increase the risk of developing lung cancer?”</i> 3. Measures: (at baseline & follow-up visits) Smoking: - Packs of cig/day <u>in the last 1 year</u> (*timeframe depends on visits) - # of years smoking - Ever/current/etc. - Type of tobacco product <u>in the last 1 year</u> Lung cancer - First diagnosis <u>in last year</u> (Y/N) - Date of physician diagnosis (MM/YY) - EHR record data of new diagnosis - Type of test to identify new diagnosis (CT)

Lesko CR et al. A Framework for Descriptive Epidemiology. AJE 2022;191(12):2063-2070.

7

Research Questions and Study Designs

2010 - 2011	2010 - 2020
(Descriptive study, e.g. Cross-sectional study) 2. Research Question: <i>“What is prevalence of LC among SMK and ~SMK across...?”</i> 3. Measures: Covariates: Does <u>prevalence</u> of smoking and lung cancer differ by: - Age - Sex/gender - Race and ethnicity - Sexual orientation - Types of discrimination (gender-, race/ethnicity-, age-based, etc.) - Geography - Other factors?	(Analytic study, e.g. Cohort Study) 2. Research Question: <i>“Does smoking increase the risk of lung cancer, adjusting for...?”</i> 3. Measures: (at baseline & follow-up visits) Covariates: Does effect of smoking on lung cancer risk remain <u>after controlling</u> for: - Age - Sex/gender - Race and ethnicity - Sexual orientation - Types of discrimination (gender-, race/ethnicity-, age-based, etc.) - Geography - Other factors? - NOTE: USE CAUSAL DIAGRAM TO IDENTIFY COVARIATES CORRECTLY

8

Target population, sampling and study designs

2010 - 2011	2010 - 2020
(Descriptive study, e.g. Cross-sectional study) 2. Research Question: <i>"What is prevalence of LC among SMK and ~SMK??"</i> 4. Target population and study sample - Does the sampling strategy employed to obtain the study sample enable us to obtain estimates that reflect the true population parameters? - <u>No counterfactual consideration!</u>	(Analytic study, e.g. Cohort Study) 2. Research Question: <i>"Does smoking increase the risk of developing lung cancer?"</i> 4. Target population and study sample - Is there a target population from which the study sample is derived? - How do we identify an 'exposed' and 'unexposed' group in the study sample? - Are the exposed and unexposed groups exchangeable in terms of their risk for developing the outcome? - Is the 'unexposed' group a good proxy for the <u>counterfactual</u> (the disease risk in the exposed had they not been exposed?)

9

Analytic plan and study designs

2010 - 2011	2010 - 2020
(Descriptive study, e.g. Cross-sectional study) 2. Research Question: <i>"What is prevalence of LC among SMK and ~SMK across...?"</i> 5. Analysis Plan - Descriptive stats on SMK - Descriptive stats on LC - Bivariable Analyses: Stratified analyses to see differences in relationship across covariate groups - *Multivariable analysis if adjustment is warranted	(Analytic study, e.g. Cohort Study) 2. Research Question: <i>"Does smoking increase the risk of developing lung cancer?"</i> 5. Analysis Plan - Descriptive stats on SMK, LC, covariates of interest - Bivariable stats to assess relationships based on causal diagram - Multivariable models to understand effect of SMK on LC, controlling for covariates OR differences by level of covariate

10



NYU

SCHOOL OF GLOBAL
PUBLIC HEALTH

Cohort studies: Design considerations

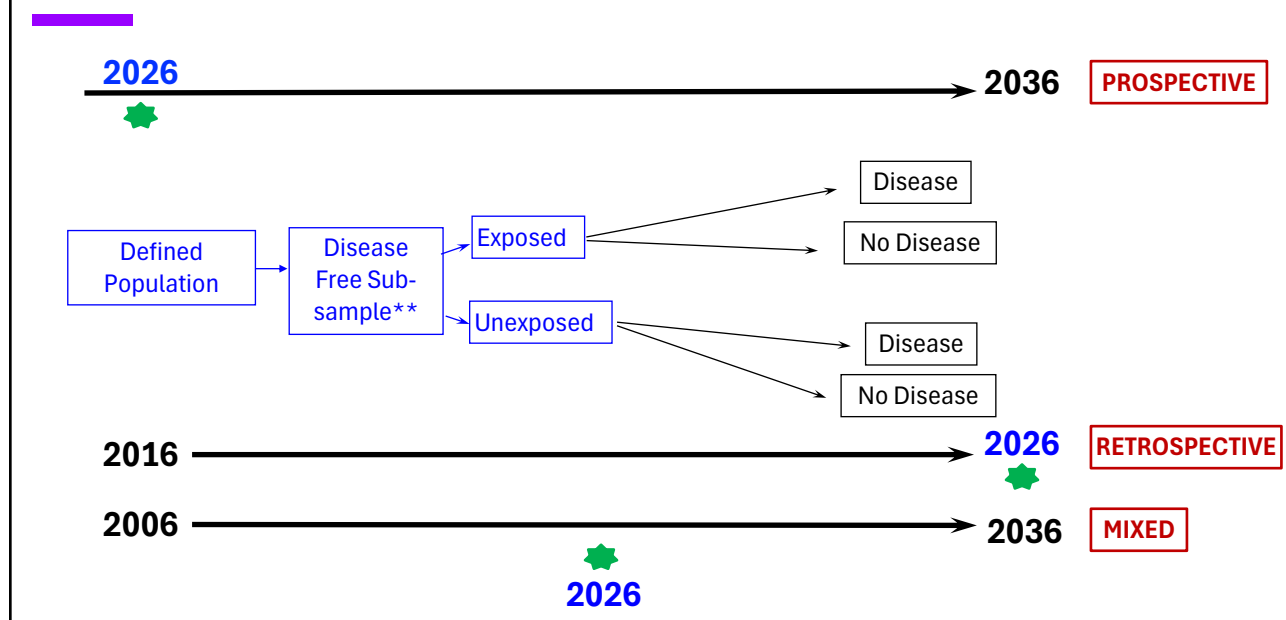
11

Cohort Study Overview

- **Traditionally:** A group of disease-free people is identified and followed up for a certain time period to ascertain the occurrence of health-related events
 - Can include individuals with and without outcome ONLY if the researcher has clearly articulated the objectives and research questions and analyses distinct for individuals with vs. without the outcome. For example:
 - Studies of disease progression
 - Studies in total populations (i.e. Framingham) where you have people with and without outcomes of interest
- Exposure status identified at baseline and assessed at follow-up visits
 - Allows for repeated-measures analysis
- Outcomes measured at baseline and at f/u
 - Definition of the outcome of interest is paramount

12

Cohort study – design types



13

Attributes of cohort studies

- Permit calculation of incidence
 - Risk
 - Rates
 - Survival
- Can be thought of as going from association to effect
- Typically: individuals form the unit of observation and the unit of analysis
- Involve the collection of primary data, although secondary data sources are used sometimes for both exposure and disease assessment

14

Deciding on population

- Defined
 - By age, sex, location, exposure, etc.
- Fixed
 - No participants are added after initial enrolment
- Dynamic
 - Cohort varies over the course of follow-up
 - Cohort can be reopened if sample size decreased due to mortality and/or attrition

15

Assembling a cohort

- Population-based
 - Includes either entire population or representative sample of population
 - Exposures unknown until first observation, when exposure information is collected
- Exposure-based
 - Find subjects with a common exposure (e.g., asbestos workers), and then identify group of individuals without exposure who are similar to exposed individuals; otherwise may have selection bias (e.g., healthy worker effect)
 - Good for rare exposures

16

Measuring exposures, covariates and outcomes

- Because a cohort study follows people over time, you can have both fixed and time-varying variables
- Fixed
 - Variables that do not vary over time (i.e., sex [not gender], blood type, etc.)
- Time-dependent
 - Outcome status (yes/no) or level
 - Exposure status or level
 - Confounders and effect modifiers may be time-dependent as well!

17

Measuring outcomes

- Single event/discrete outcome
 - Observe to the event, follow-up (FU) ends at event
 - Event can only happen once
- Multiple events/discrete outcome
 - Observe first event; accrual of FU begins after end of 1st event
- Intermediate markers of outcomes
 - Rate of change in a measure predictive of outcome (i.e., rate of decline of some measure)
 - Outcome that represents an intermediate step in disease process

18

Exchangeability in cohort studies

- We want exposed and unexposed persons as comparable as possible – we want them to be **exchangeable**
- But exposure is self-determined – no randomization
- Selection into study can vary by exposure group → unexposed group may not be good proxy for exposed group had they not been exposed
- In order to achieve exchangeability and come close to achieving the ideal, we need to address challenges in observational studies
 - Essentially: we **must** deal with confounding.

19

Cohort study design considerations

- Cohort population
 - Overall
 - Exposure opportunity
 - Outcome risk
- Participant enrollment
- Defining & assessing
 - exposures
 - covariates
- Subject follow-up and retention
- Defining & assessing outcomes

20

Cohort population, overall

- Overall, who is included and why?
 - Identify a relevant population
 - What makes a population relevant for a study?

21

Cohort population: exposure opportunity

- Exposure opportunity
 - People should be included who could get the exposure, at least in theory.
 - Don't study the effect of hysterectomy in people who don't (and have never had) a uterus: their probability of being exposed to hysterectomy is zero.
- Of course, this is true in an experiment, as well.
 - If you were randomizing to penicillin vs. tetracycline in a trial, people with penicillin allergy would be ineligible
 - Can be difficult to know in observational setting, if you are not collecting exposure data
 - For example: If using healthcare database, if some people get tetracycline bc of penicillin allergy may not be known

22

Cohort population: outcome risk

- Include people free of outcome
- Only include people at risk of the outcome.
 - Would we study risk of uterine cancer among women who have had hysterectomy?
 - More “immortal person-time” can be introduced this way
- We therefore collect prospective incident cases in a group that is initially free of (but at risk for!) disease of interest, and possibly who are free of exposure at baseline (but not necessarily!).
- We also want a group which is likely to experience the disease frequently enough (for the size of the cohort) that we have sufficient power.

23

Participant enrollment

- Finding participants: where? How selected?
- Ethical issues
 - Still exist, even though you’re not assigning intervention status
 - Some of the worst ethical violations in recent history were in the setting of an observational study: **Tuskegee Syphilis Study (1932–1972)**
- Obtaining consent
 - Can be complicated
 - May not be necessary if data are already collected. E.g., using de-identified data

24

Defining exposure and covariates

- Exposures:
 - A well-defined study question
 - Then need to consider whether exposure is measured well
 - Again, consider how the observational nature of the study affects consistency in ascertainment and collection of exposure data
- Covariates:
 - How do we decide which covariates are worth collecting?
 - Again, pointing forward to confounding.
- In general, we mostly worry about exposure and outcome; but measurement of covariates is also vital

25

Assessment of exposure and covariates

- Often assessed at baseline (cohort entry)
- Questionnaires should be “exhaustive but not exhausting”
- Try very, very hard to eliminate missing data, as it can lead to bias and will reduce power.
- What happens when exposures (and covariates) change
 - For example, in a cohort of HIV-positive people who are enrolled at seroconversion, the exposure of use of antiretroviral therapy changes over time.
 - Continued assessment of exposure and covariates may be necessary and greatly complicates analysis.

26

Participant follow-up & retention

- Follow-up: Tracking all participants forward in time
 - How often should you track them?
 - Are you going to miss events, or exposures?
 - Tradeoffs between costs and effort, and coarseness of data
- Retention:
 - Good retention is critical!
 - Many and different attempts to make contact
 - E.g.: multiple mailings; phone calls at varying times of day; reminders via smartphone
 - Providing significant benefits to participants may help keep them coming back
 - But there are ethical challenges → remuneration vs. coercion

27

Defining and assessing outcomes

- A well-articulated study question makes this easier!
 - Who got sick? How is the outcome defined and measured?
- Constant monitoring for key outcomes is desirable, but not always possible.
 - Type of cohort population may impact this!
- A planned, formal, interval cohort like the WHS (come back every six months for f/u visits) will miss the actual dates of most important events
 - I can tell you that I had a heart attack or a new cancer diagnosis since the last visit. I may not be able to accurately tell you exactly when that was.
 - For less critical events (how many times have you had an upper respiratory infection in the last six months?) recall may be even worse, and count may be wrong.
 - This may result in misclassification of outcome time, number, severity...and a thing we call interval censoring.
 - More frequent follow-up may be desirable for less-critical events.

28

Defining and assessing outcomes (cont.)

- Loss to follow-up will prevent outcome assessment
- Why do people go LTFU?
 - Subject withdraws from study
 - Subject cannot be located
 - Why? Emigration; changed contact information
 - Cell phone numbers change very frequently in some places.
- Loss to follow-up creates missing data and potential for selection bias
- For certain outcomes in certain settings, external data sources (death registry, cancer registry) are possible, and can eliminate some of these problems.